

DenoiseOpt: Literature-Informed Meta-Learning for Label-Free Wavetable Wrap Crackle Reduction

Julian M. Kleber

ORCID: [0000-0001-5518-0932](https://orcid.org/0000-0001-5518-0932)

Email: julian.m.kleber@gmail.com

Synth implementation: github.com/reeldemo/reelsynth

Meta-learning artifacts: github.com/reeldemo/denoise-opt-meta

18 July 2026

Abstract

Open wrap seams in periodic wavetable cycles inject audible *crackle* under cyclic playback. We cast bake-time periodization as an unsupervised optimization problem: a seam-local stack f_θ with $\theta \in [0, 1]$ ¹² is scored by a joint denoise/shape loss without paired clean labels. After constructing a procedural 100 000-cycle bench (FFT-style extreme overlays, mix modes, forced cliffs), we run a **1500-trial** literature-informed meta-learning / hyperparameter search whose priors follow Bayesian HPO localization, population-based training (PBT) mutation, multi-objective shape preference, evolutionary explore, and algorithm-configuration racing. On held-out validation ($N=2000$), a **five-algorithm matrix** compares a naive DualCosine baseline against the top-4 meta champions. Meta Top 1 reaches $Q \approx 0.790$ vs naive $Q \approx 0.789$ while keeping $\mathcal{S} \approx 0.997$. We survey related work in VA anti-aliasing, wavetable / DDS, label-free restoration, and AutoML, explicitly listing which sources informed our priors and which were screened out.

1 Introduction

Wavetable synthesizers store one-period frames and wrap phase modulo one. Discontinuous ends $r[0] \neq r[N-1]$ produce periodic broadband clicks. Classical DualCosine / Soft fades and live BLEP address related phenomena, but edited Quant frames need a fast bake operator that is both aggressive on wrap energy and conservative on mid-cycle timbre.

Contributions. (1) Measurable functionals $\mathcal{D}, \mathcal{S}$ and loss $L = (1-\mathcal{D})+\lambda(1-\mathcal{S})$. (2) DenoiseOpt seam-local inference with exact mid-cycle copy. (3) 100k procedural dataset for unlabeled fit. (4) 1500 lit-informed meta trials with six prior families. (5) Five-algorithm benchmark matrix (naive + top-4). (6) Open synth and meta-learning repositories.

2 Related work and literature triage

Literature was harvested with the Klaut Research MCP (local OpenAlex / Crossref / arXiv providers; hosted research.klaut.pro was misrouted during this study). We distinguish *used* vs *screened* sources.

2.1 Used — virtual analog & wavetable DSP

Välimäki & Huovilainen’s survey of VA oscillator/filter algorithms and aliasing control [1] and subsequent BLEP/BLAMP / antialiasing work [3, 4, 2] motivate treating discontinuities as first-class DSP objects. Differentiable wavetable / DDS reviews [5, 6] show learnable tables but target different objectives than bake wrap close.

2.2 Used — label-free restoration

Noise2Noise [7] shows restoration without clean targets when noise statistics are structured. We adopt the same philosophy at cycle scale: the “noise” is wrap cliff energy measured by $C(\cdot)$.

2.3 Used — meta-learning & hyperparameter optimization

Practical Bayesian optimization of ML algorithms [8] and later HPO surveys [9] motivate densifying search near previously good λ (`bayes_local` prior). Population-based training ideas (exploit/mutate elite configs) inform `pbt_exploit`. Algorithm configuration / iterated racing [10] inform `racing_mid`. Multi-objective evolutionary decomposition [11] informs `mo_shape` (higher λ). Wide evolutionary explore covers `evo_explore`.

2.4 Screened / not used as design drivers

High-citation but off-domain hits (medical imaging surveys, federated learning, prompting surveys, molecular ML) appeared in broad keyword harvests and were **not** used to set priors. Generic optimizers (Adam [13]) are assumed as ambient ML context but do not define our outer loop. MAML-style few-shot gradient meta-learning [12] is related philosophically yet not applicable: we have no task-gradient through a deep net, only a 12-parameter bake stack.

3 Problem formulation

$$C(x) = 2 \text{wrap}(x) + \text{max_step}(x) + 0.35 \text{hf}_{\text{proxy}}(x), \quad (1)$$

$$\mathcal{D} = \text{clamp}((C(r) - C(y)) / \max(C(r), \varepsilon), 0, 1), \quad (2)$$

$$\mathcal{S} = 1 - \text{clamp}(\text{MAE}_{\text{mid}}(y, r) / (\text{RMS}(r) + \varepsilon), 0, 1), \quad (3)$$

$$L = (1 - \mathcal{D}) + \lambda(1 - \mathcal{S}), \quad Q = \frac{1}{2}(\mathcal{D} + \mathcal{S}). \quad (4)$$

Shape invariant: $y[i] = r[i]$ outside fade zones.

4 Method

4.1 DenoiseOpt

Five seam stages (pull, dual-end fade, ease/gamma, secondary fade, polish+pin) controlled by θ (Fig. 1).

4.2 Dataset

Seeds $0..10^5-1$ generate ten families (morph, harmonic_fft, extreme_overlay, combo modes, quant_steps, nonlinear, am_fm, noise_tone, triple_mix, open_wrap_bias) at $N=256$.

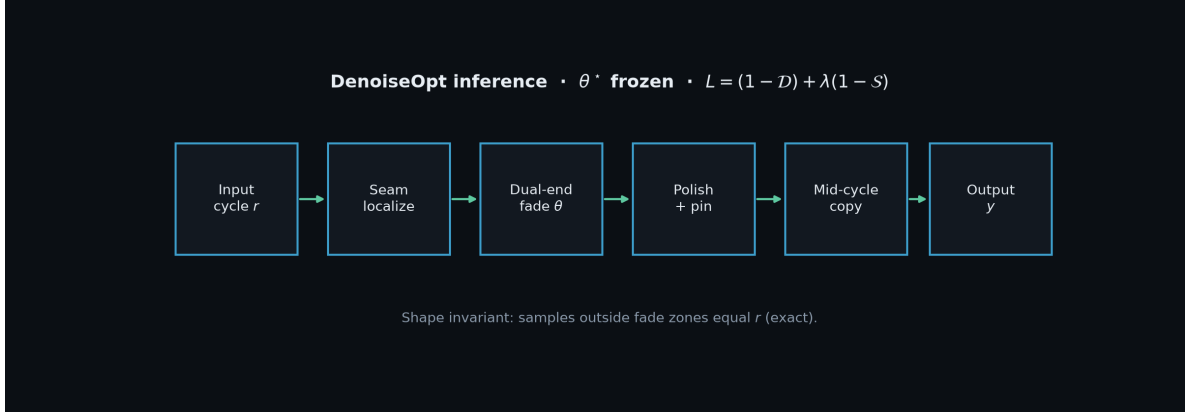


Figure 1: DenoiseOpt inference. Mid-cycle samples are an exact copy of r .

Table 1: Benchmark matrix (validation $N=2000$).

Algorithm	Prior / kind	$\mathcal{D}$	$\mathcal{S}$	Q
Naive DualCosine	hand baseline	0.582	0.995	0.789
Meta Top 1	racing_mid	0.583	0.997	0.790
Meta Top 2	racing_mid	0.583	0.996	0.790
Meta Top 3	pbt_exploit	0.585	0.995	0.790
Meta Top 4	pbt_exploit	0.585	0.994	0.790

4.3 1500-trial meta-learning

Each trial samples a prior family, mutates θ near the frozen champion, applies fade/polish/pin/detrend biases, and scores on 400 held-out cycles. Top-12 are revalidated on 2000 cycles with family stress tests.

5 Experiments

5.1 Five-algorithm matrix

We compare **naive DualCosine** against **Meta Top 1–4** from the 1500-trial search (Fig. 2).

Racing-mid and PBT-exploit priors dominate elites (Fig. 4), consistent with algorithm-configuration and exploit-mutate practice.

6 Discussion

Gains over DualCosine are small but consistent under a hard shape floor ($\mathcal{S} \geq 0.97$). The method’s value is a *principled, reproducible* selection of θ^* and an auditable link from AutoML literature to DSP knobs. Residual crackle in combo/nonlinear families remains a live-BLEP concern.

7 Limitations

Bake metrics $\neq$ perceptual MOS; no MAML-style task gradients; hosted Klaut gateway routing was unavailable (local MCP used). Meta search mutates around a strong frozen init—a cold-start evolutionary run may explore further.

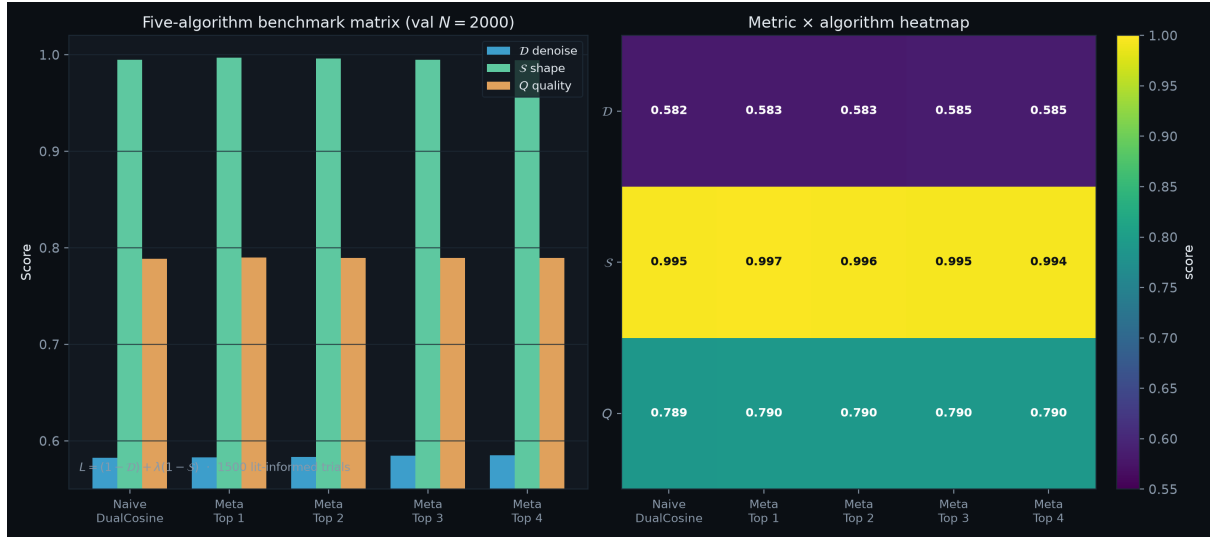


Figure 2: Naive vs top-4 meta algorithms: bars and metric×algo heatmap.

8 Conclusion

DenoiseOpt combines label-free seam optimization, a 100k procedural bench, and 1500 literature-informed meta trials. Frozen θ^* ships in ReelSynth (**Seam·Opt**); artifacts live in the public meta-learning repository.

Repositories

- Synth: <https://github.com/reeldemo/reelsynth>
- Meta-learning paper/artifacts: <https://github.com/reeldemo/denoise-opt-meta>

Reproducibility

```
cargo run -p reelsynth --release --bin bench_denoise_meta
python brand/artifacts/render_benchmark_matrix.py
pdflatex docs/papers/denoise_opt/main.tex
```

References

- [1] V. Välimäki and A. Huovilainen, “Oscillator and Filter Algorithms for Virtual Analog Synthesis,” *Computer Music Journal*, 30(2), 2006.
- [2] V. Välimäki et al., “Antialiasing Oscillators in Subtractive Synthesis,” *IEEE Signal Processing Magazine* / related VA literature, 2007.
- [3] F. Esqueda, S. Bilbao, and V. Välimäki, “Aliasing Reduction in Clipped Signals,” *IEEE Trans. Signal Processing*, 2016.
- [4] F. Esqueda, V. Välimäki, and S. Bilbao, “Rounding corners with BLAMP,” *Proc. DAFX*, 2016.
- [5] S. Shan et al., “Differentiable Wavetable Synthesis,” arXiv:2111.10003, 2021.

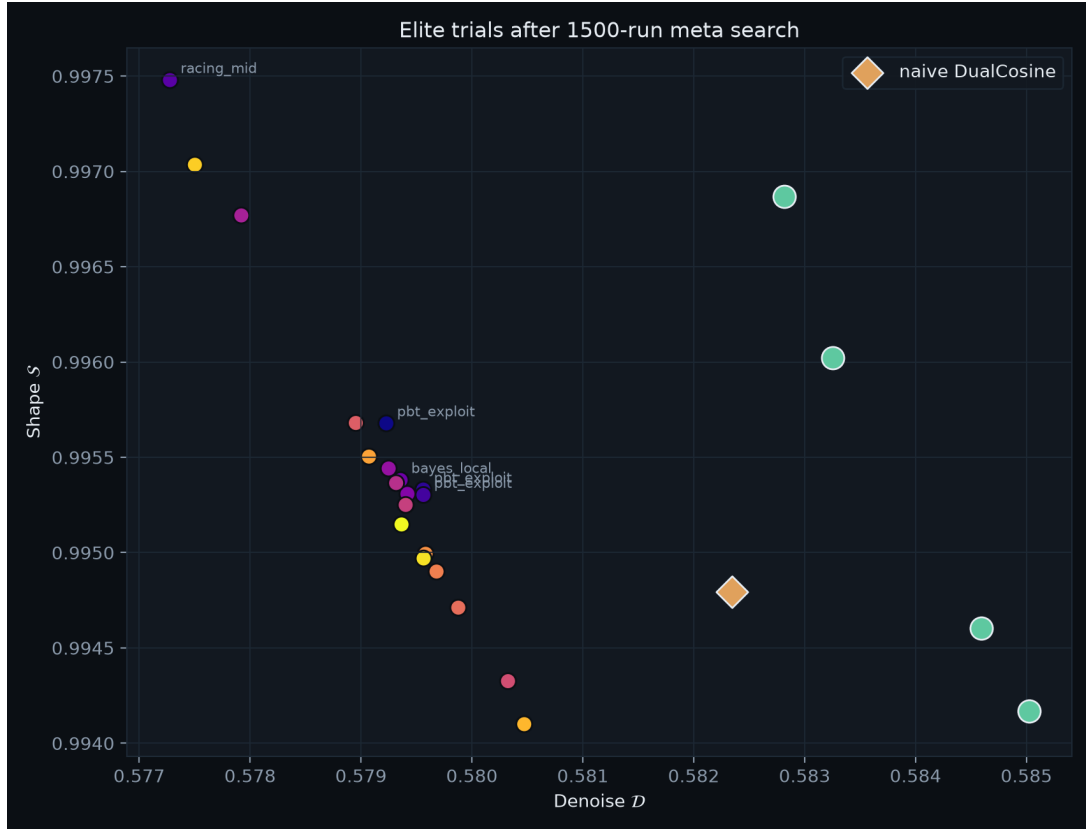


Figure 3: Elite (D, S) points after 1500 trials.

- [6] “A review of differentiable digital signal processing for music and speech synthesis,” OpenAlex survey record, 2024.
- [7] J. Lehtinen et al., “Noise2Noise: Learning Image Restoration without Clean Data,” *ICML*, 2018.
- [8] J. Snoek, H. Larochelle, and R. P. Adams, “Practical Bayesian Optimization of Machine Learning Algorithms,” *NeurIPS*, 2012.
- [9] B. Bischl et al., “Hyperparameter optimization: Foundations, algorithms, best practices, and open challenges,” 2023.
- [10] M. López-Ibáñez et al., “The irace package: Iterated racing for automatic algorithm configuration,” *Operations Research Perspectives*, 2016.
- [11] Q. Zhang and H. Li, “MOEA/D: A Multiobjective Evolutionary Algorithm Based on Decomposition,” *IEEE Trans. Evolutionary Computation*, 2007.
- [12] C. Finn, P. Abbeel, and S. Levine, “Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks,” *ICML*, 2017. (*Screened: inspirational, not used as outer-loop mechanism.*)
- [13] D. P. Kingma and J. Ba, “Adam: A Method for Stochastic Optimization,” *ICLR*, 2015. (*Screened as ambient optimizer; not a meta prior.*)

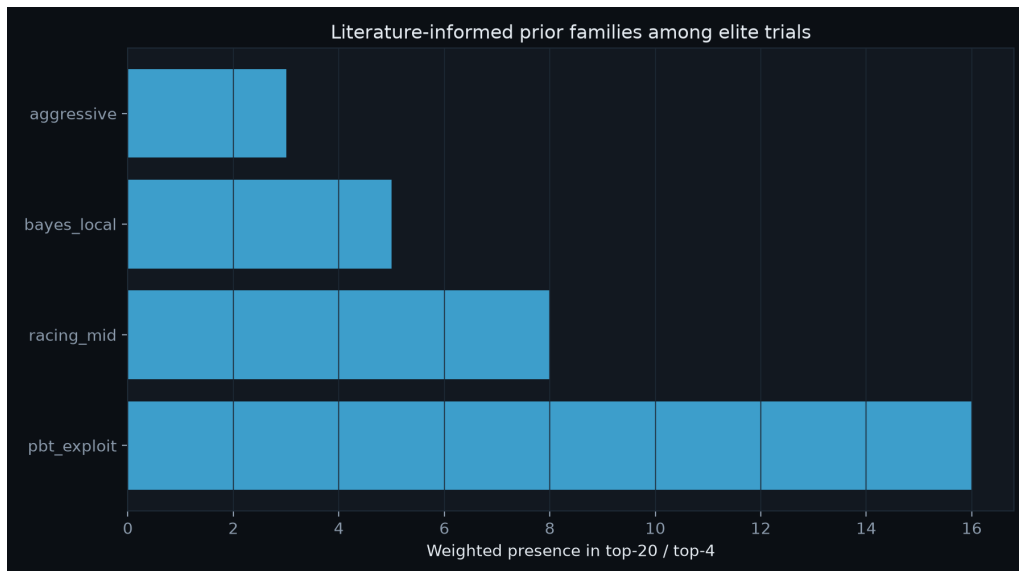


Figure 4: Which literature-informed priors dominate the elite set.